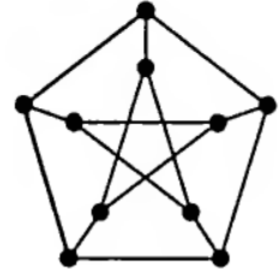


Exercise C1

- (a) Draw the six unlabelled trees with six vertices.
- (b) By adding a new edge in all possible ways to each unlabelled tree with six vertices, draw the eleven unlabelled trees with seven vertices.
- (c) Give an example of a tree with seven vertices and
 - (1) exactly two vertices of degree 1;
 - (2) exactly four vertices of degree 1;
 - (3) exactly six vertices of degree 1.
- (d) Find three spanning trees in the Petersen graph.
- (e) How many spanning trees has $K_{2,3}$? $K_{2,100}$?
- (f) Let G be a connected graph. What can you say about
 - (1) an edge of G that appears in every spanning tree?
 - (2) an edge of G that appears in no spanning tree?



Exercise C2

- (a) Prove that every tree is a bipartite graph.
- (b) Which trees are complete bipartite graphs?
- (c) Use a proof by contradiction to show that the removal of an edge cannot disconnect a tree into more than two components.

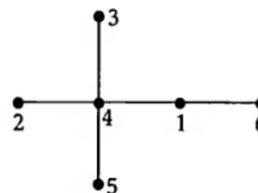
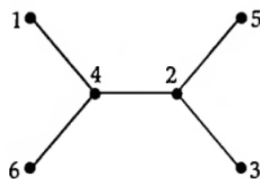
Exercise C4

A forest is a graph (not necessarily connected), each of whose components is a tree.

- (a) Let G be a forest with 11 vertices and k components. How many edges does G have?
- (b) Construct a forest with 12 vertices and 9 edges.
- (c) Is it true that every forest with k components has at least $2k$ vertices of degree 1?

Exercise C5

- (a) Write down the Prufer sequence associated with each of the following labelled trees.



- (b) Draw the labelled tree associated with each of the Prufer sequences (1, 2, 3, 4) and (3, 3, 3, 3).

Exercise P1

- (a) Find the number of trees on n vertices in which a given vertex is an end-vertex.
- (b) Deduce that, if n is large, then the probability that a given vertex of a tree with n vertices is an end-vertex is approximately e^{-1} .

Exercise P2

Provide a proof that the graph resulting from the Prufer sequence corresponds to a tree, implying it is both connected and acyclic.

Exercise P3

Prove that every n -vertex graph with m edges has at least $m - n + 1$ cycles.

Exercise P4

Let $d_1, d_2, \dots, d_n$ be positive integers, with $n \geq 2$. Prove that there exists a tree with vertex degrees $d_1, d_2, \dots, d_n$ if and only if $\sum d_i = 2n - 2$.

Exercise P5

Let T be a tree in which all vertices adjacent to leaves have degree at least 3.

Prove that T has some pair of leaves with a common neighbor.

Exercise P6

A center of G is a vertex u such that $\max d(u, v)$ is as small as possible. Show that a tree has either exactly one center or two, adjacent, centers.

Exercise P7

Prove that every n -vertex tree other than $K_{1,n-1}$ is contained in its complement. (Hint: Use induction on n to prove a stronger statement: if T is an n -vertex tree other than a star, then K_n contains two edge-disjoint copies of T in which the two copies of each non-leaf vertex of T appear at distinct vertices.) (Bums-Schuster [1978])

Exercise P8

Use Cayley's Formula to prove that the graph obtained from K_n by deleting an edge has $(n-2)n^{n-3}$ spanning trees.